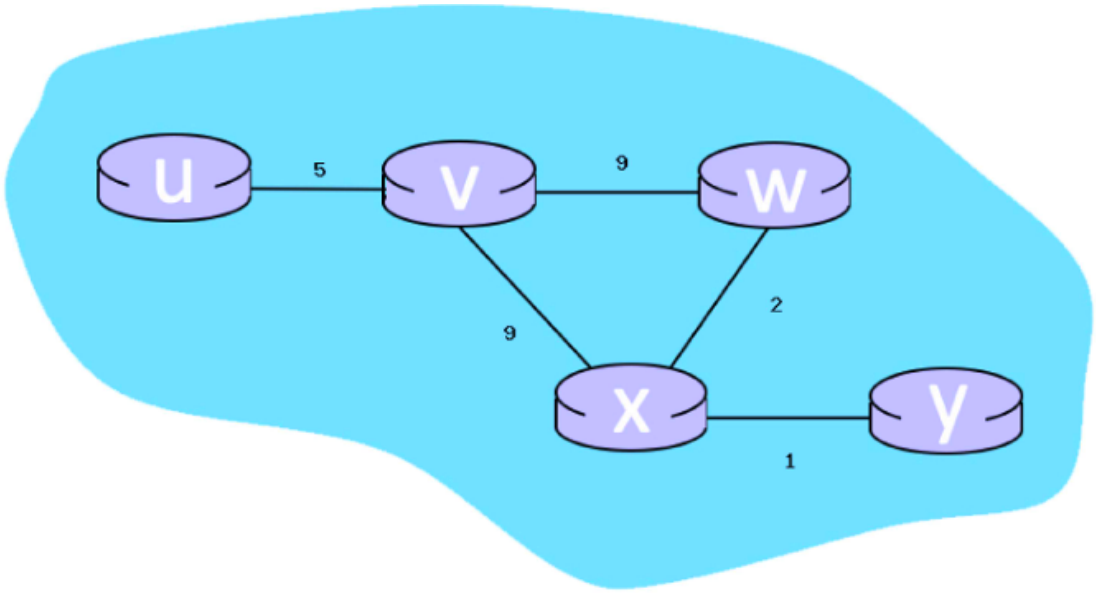


QUESTION LIST

1. When the algorithm converges, what are the distance vectors from router 'X' to all routers? Write your answer as u,v,w,x,y

2. What are the initial distance vectors for router 'V'? Write your answer as u,v,w,x,y and if a distance is ∞, write 'x'

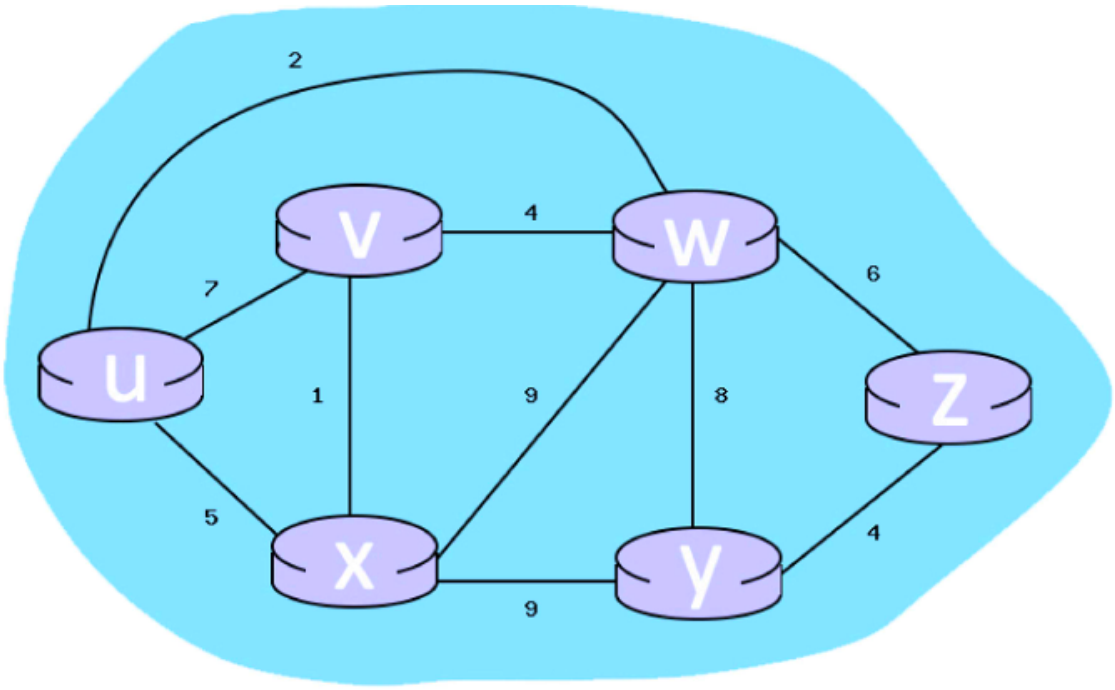
3. The phrase 'Good news travels fast' is very applicable to distance vector routing when link costs decrease; what is the name of the problem that can occur when link costs increase?

SOLUTION

1. When the algorithm converges, router X has distance vectors (u,v,w,x,y) = (14,9,2,0,1)

2. The initial distance vectors of router V are: (u,v,w,x,y) = (5,0,9,9,x) where x is ∞

3. It is called the 'Count to Infinity' problem.



Using Dijkstra's algorithm, find the least cost path from source node U to all other destinations and answer the following questions

QUESTION LIST

1. What is the shortest distance to node z and what node is its predecessor? Write your answer as n,p

2. What is the shortest distance to node y and what node is its predecessor? Write your answer as n,p

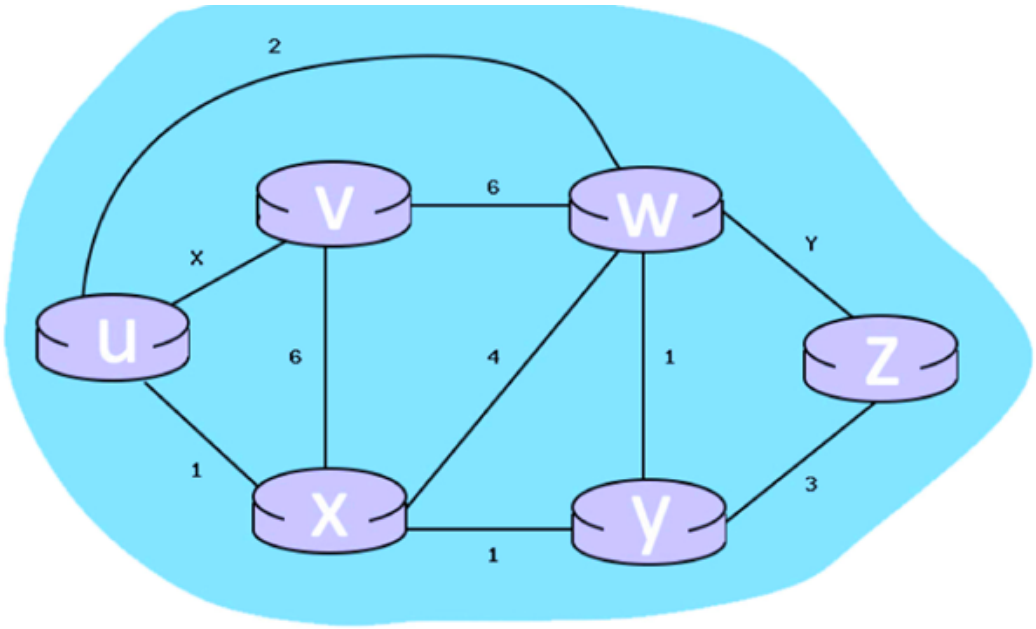
3. What is the shortest distance to node w and what node is its predecessor? Write your answer as n,p

SOLUTION

1. The minimum distance from node u to node z is 8, and node z's predecessor is node w. The full answer was: 8,w

2. The minimum distance from node u to node y is 10, and node y's predecessor is node w. The full answer was: 10,w

3. The minimum distance from node u to node w is 2, and node w's predecessor is node u. The full answer was: 2,u



Consider the completed table below, which calculates the shortest distance to all nodes from X:

Node	Shortest distance from X	Previous Node
X	0	n/a
U	1	X
Y	1	X
V	2	U
W	2	Y
Z	4	Y

QUESTION LIST

1. For link X, what is the cost associated with this link? If the answer can't be determined given the information, respond with 'n/a'

2. For link Y, what is the cost associated with this link? If the answer can't be determined given the information, respond with 'n/a'

SOLUTION

1. The prior node in the path to V is U, and we know the shortest distance of both V (2) and U (1), so 2 - 1 = 1 which is X.

2. Because the link is never used, we cannot determine the value of Y, so the answer is n/a.

ERROR DETECTION AND CORRECTION: TWO DIMENSIONAL PARITY

Suppose that a packet's payload consists of 10 eight-bit values (e.g., representing ten ASCII-encoded characters) shown below. (Here, we have arranged the ten eight-bit values as five sixteen-bit values):

Figure 1

10011110 10100000
01110100 11100011
10011010 00100010
01001101 10110001
01100001 01010100

Figure 2

Both the payload and parity bits are shown. One of these bits is flipped.

11011111 10101000 0
10101001 00010111 0
11110110 00110100 1
00011000 01000100 0
01010010 11011011 1
10001010 00010100 0

Figure 3

Both the payload and parity bits are shown; Either one or two of the bits have been flipped.

01001101 00100100 1
00111000 10010000 1
11110011 11001011 1
00111110 00000110 1
11011111 01111101 1
01100010 00000100 1

SOLUTION

The full solution for figure 1 is shown below:

10011110 10100000 1
01110100 11100011 1
10011010 00100010 0
01001101 10110001 0
01100001 01010100 0
01011100 10000100 0

1. The parity bits for the 16 columns is: 01011100 10000100

2. The parity bits for the 5 rows is: 11000

3. The parity bit for the parity row is: 0

4. The bit that was flipped in figure 2 is (1,5):

11011111 10101000 0
10101001 00010111 0
11110110 00110100 1
00011000 01000100 0
01010010 11011011 1
10001010 00010100 0

For figure 3, the bits that were flipped are (7,5) and (5,0):

01001101 00100100 1
00111000 10010000 1
11110011 11001011 1
00111110 00000110 1
11011111 01111101 1
01100010 00000100 1

5. No, with 2D parity, you can detect the presence of two flipped bits, but you can't know their exact locations in order to correct them.

ERROR DETECTION AND CORRECTION: CYCLIC REDUNDANCY CHECK

Consider the Cyclic Redundancy Check (CRC) algorithm discussed in Section 6.2.3 of the text. Suppose that the 4-bit generator (G) is 1001, that the data payload (D) is 10011100 and that r = 3.

SOLUTION

To compute the CRC, we begin by taking the value of D, 10011100, and multiplying it by 2^3, giving 10011100000. We then divide this number by the generator bits [G] = 1001, using modulo-2 arithmetic. The final remainder, R, after this division are then the CRC bits. Here is that calculation:

```
        10001101
      -------------
1001 ) 10011100000
       1001
       ----
           1100
           1001
           ----
            1010
            1001
            ----
              1100
              1001
              ----
               101------>R
```

We've computed the remainder as R = 101 and the quotient n = 10001101. You should verify that n*G XOR R is indeed equal to 10011100000. You can use this calculator to do the modulo-2 arithmetic if you don't want to do it by hand.

RANDOM ACCESS PROTOCOLS: ALOHA

Assume that there are 2 active nodes, each of which has an infinite supply of frames they want to transmit, and these frames have a constant size of L bits. If two or more frames collide, then all nodes will detect the collision.

There are two versions of the Aloha protocol: Slotted and Pure. In this problem we will be looking at the efficiency of these two variations. In the case of Slotted Aloha, frames will be sent only at the beginning of a time slot, frames take an entire time slot to send, and the clocks of all nodes are synchronized.

Please round all answers to 2 decimal places.

SOLUTION

1. The efficiency given p = 0.22 is: Np(1 - p)^2(N - 1) = 2 * 0.22 * (1 - 0.22)^2(2 - 1) = 0.27 or 27% efficiency.

2. The efficiency given p = 0.5 is: Np(1 - p)^2(N - 1) = 2 * 0.5 * (1 - 0.5)^2(2 - 1) = 0.25 or 25% efficiency.

RANDOM ACCESS PROTOCOLS: ALOHA

Assume that there are 2 active nodes, each of which has an infinite supply of frames they want to transmit, and these frames have a constant size of L bits. If two or more frames collide, then all nodes will detect the collision.

There are two versions of the Aloha protocol: Slotted and Pure. In this problem we will be looking at the efficiency of these two variations. In the case of Slotted Aloha, frames will be sent only at the beginning of a time slot, frames take an entire time slot to send, and the clocks of all nodes are synchronized.

Please round all answers to 2 decimal places.

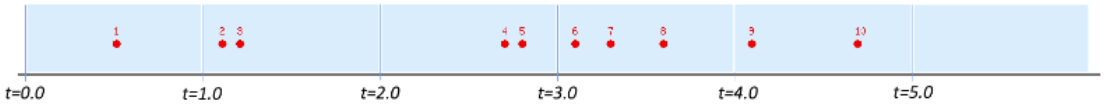
SOLUTION

1. The efficiency given p = 0.22 is: Np(1 - p)^(N - 1) = 2 * 0.22 * (1 - 0.22)^(2 - 1) = 0.34 or 34% efficiency.

2. The efficiency given p = 0.5 is: Np(1 - p)^(N - 1) = 2 * 0.5 * (1 - 0.5)^(2 - 1) = 0.5 or 50% efficiency.

MULTIPLE ACCESS PROTOCOLS: COLLISIONS

Consider the figure below, which shows the arrival of 10 messages for transmission at different multiple access wireless nodes at times t = <0.5, 1.1, 1.2, 2.7, 2.8, 3.1, 3.3, 3.6, 4.1, 4.7> and each transmission requires exactly one time unit.



QUESTION LIST

1. Suppose all nodes are implementing the Aloha protocol. For each message, indicate the time at which each transmission begins. Separate each value with a comma and no spaces.

2. Which messages transmit successfully? Write your answer as a comma seperated list with no spaces using the messages' numbers

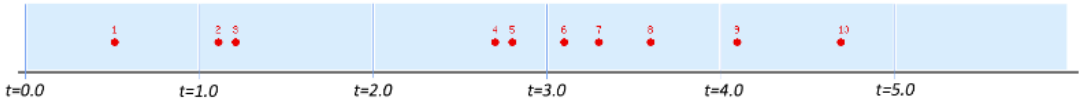
SOLUTION

1. The list of times for frame transmissions is: 0.5,1.1,1.2,2.7,2.8,3.1,3.3,3.6,4.1,4.7

2. The list of successful frames is:

MULTIPLE ACCESS PROTOCOLS: COLLISIONS

Consider the figure below, which shows the arrival of 10 messages for transmission at different multiple access wireless nodes at times t = <0.5, 1.1, 1.2, 2.7, 2.8, 3.1, 3.3, 3.6, 4.1, 4.7> and each transmission requires exactly one time unit.



QUESTION LIST

1. Suppose all nodes are implementing the Slotted Aloha protocol. For each message, indicate the time at which each transmission begins. Separate each value with a comma and no spaces.

2. Which messages transmit successfully? Write your answer as a comma seperated list with no spaces using the messages' numbers

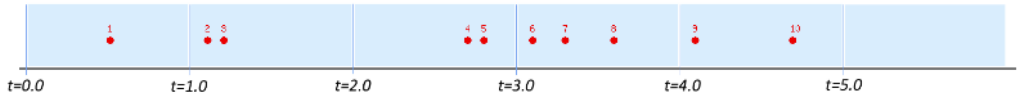
SOLUTION

1. The list of times for frame transmissions is: 1,2,2,3,3,4,4,4,5,5

2. The list of successful frames is: 1

MULTIPLE ACCESS PROTOCOLS: COLLISIONS

Consider the figure below, which shows the arrival of 10 messages for transmission at different multiple access wireless nodes at times t = <0.5, 1.1, 1.2, 2.7, 2.8, 3.1, 3.3, 3.6, 4.1, 4.7> and each transmission requires exactly one time unit.



QUESTION LIST

1. Suppose all nodes are implementing Carrier Sense Multiple Access (CSMA), but without collision detection. Suppose that the time from when a message transmission begins until it is beginning to be received at other nodes is 0.4 time units. (Thus if a node begins transmitting a message at t=2.0 and transmits that message until t=3.0, then any node performing carrier sensing in the interval [2.4, 3.4] will sense the channel busy.) For each message, indicate the time at which each message transmission begins, or indicate that message transmission does not begin due to a channel that is sensed busy when that message arrives. Separate each value with a comma and no spaces, and if the channel is sensed busy, substitute it with 's'

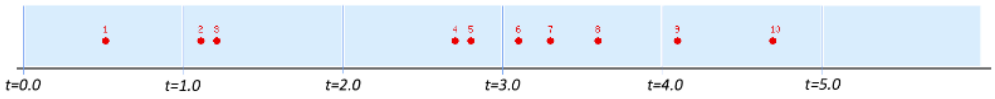
2. Which messages transmitted successfully? Write your answer as a comma seperated list with no spaces using the messages' numbers

SOLUTION

1. The list of times for frame transmissions is: 0.5,s,s,2.7,2.8,s,s,s,s,4.7

2. The list of successful frames is: 1,10

Consider the figure below, which shows the arrival of 10 messages for transmission at different multiple access wireless nodes at times t = <0.5, 1.1, 1.2, 2.7, 2.8, 3.1, 3.3, 3.6, 4.1, 4.7> and each transmission requires exactly one time unit.



QUESTION LIST

1. Suppose all nodes are implementing Carrier Sense Multiple Access (CSMA), with collision detection (CSMA/CD). Suppose that the time from when a message transmission begins until it is beginning to be received at other nodes is 0.4 time units, and assume that a node can stop transmission instantaneously when a message collision is detected. (Thus if a node begins transmitting a message at t=2.0 and transmits that message until t=3.0, then any node performing carrier sensing in the interval [2.4, 3.4] will sense the channel busy.) For each message, indicate the time at which each message transmission begins, or indicate that message transmission does not begin due to a channel that is sensed busy when that message arrives. Separate each value with a comma and no spaces, and if the channel is sensed busy, substitute it with 's'

2. Which messages transmitted successfully? Write your answer as a comma seperated list with no spaces using the messages' numbers

3. At what time did each message stop transmitting due to a collision. Write your answer as a comma seperated list with no spaces using the messages' numbers in order, and if a message didn't stop, write 'x' for that message

SOLUTION

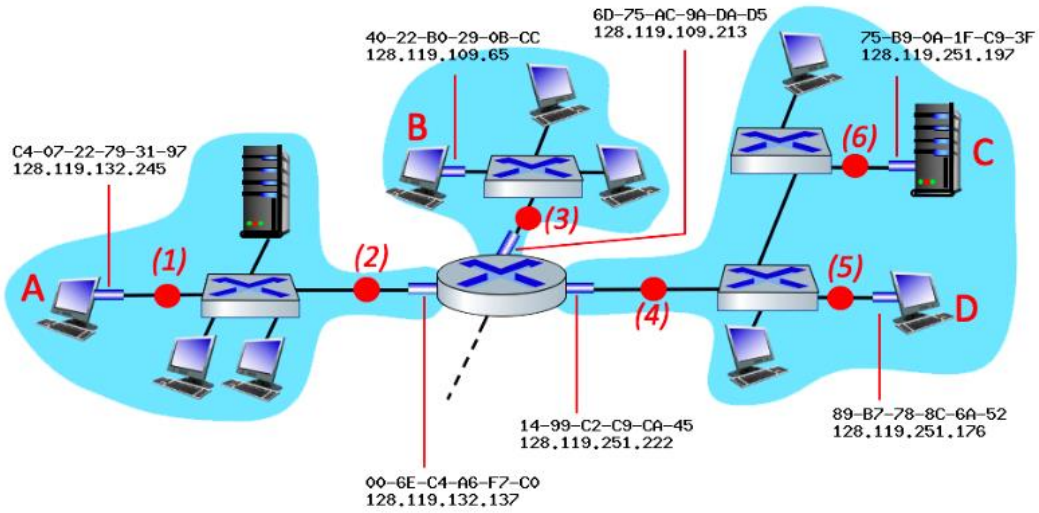
1. The list of times for frame transmissions is: 0.5,s,s,2.7,2.8,s,s,3.6,s,s

2. The list of successful frames is: 1,8

3. The list of stopped packet times is: x,x,x,3.2,3.1,x,x,x,x,x

LINK LAYER (AND NETWORK LAYER) ADDRESSING AND FORWARDING

Consider the figure below. The IP and MAC addresses are shown for nodes A, B, C and D, as well as for the router's interfaces.



Consider an IP datagram being sent from node C to node B.

QUESTION LIST

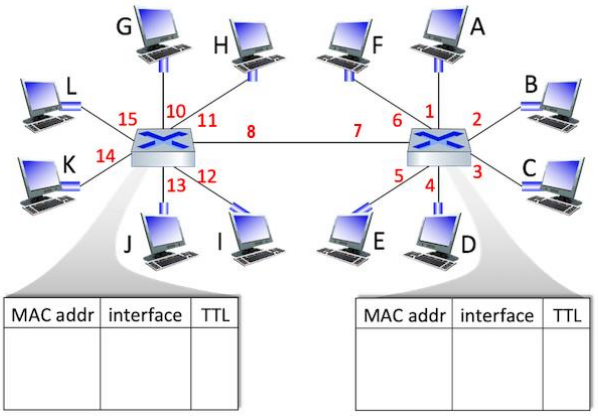
1. What is the source mac address at point 6?
2. What is the destination mac address at point 6?
3. What is the source IP address at point 6?
4. What is the destination IP address at point 6?
5. Do the source and destinaton mac addresses change at point 4? Answer with yes or no.
6. Do the source and destinaton mac addresses change at point 3? Answer with yes or no.
7. What is the source mac address at point 3?
8. What is the destination mac address at point 3?

SOLUTION

1. The source mac address at point 6 is 75-B9-0A-1F-C9-3F
2. The destination mac address at point 6 is 14-99-C2-C9-CA-45
3. The source IP address at point 6 is 128.119.251.197
4. The destination IP address at point 6 is 128.119.109.65
5. No, datagrams can be sent across the subnet via the link layer in one go.
6. Yes, the datagram is being passed through a router, which causes a subnet change.
7. The source mac address at point 3 is 6D-75-AC-9A-DA-D5
8. The destination mac address at point 3 is 40-22-B0-29-0B-CC

LEARNING SWITCHES - BASIC

Consider the LAN below consisting of 10 computers connected by two self-learning Ethernet switches. (You may want to re-read section 6.4.3 in the 8th edition textbook). At t=0 the switch table entries for both switches are empty. At t = 1, 2, 3, and 4, a source node sends to a destination node as shown below, and the destination replies immediately (well before the next time step).



Assume that the following transmissions occur (the transmissions in reply occur but are not shown in the list below):

t=1: B -> A
t=2: K -> B
t=3: K -> H
t=4: I -> F

QUESTION LIST

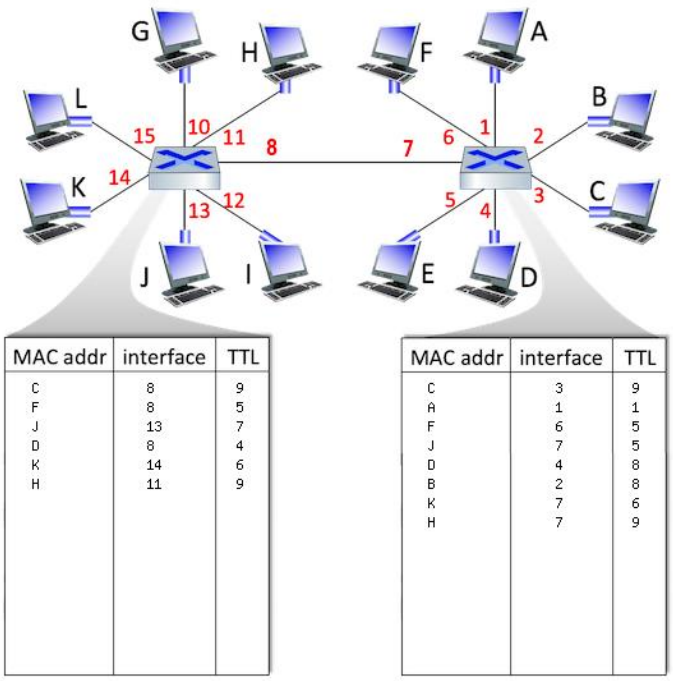
1. At t=1, what is the source entry for switch 1? Format your answer as letter,number or 'n/a'
2. At t=1, what is the destination entry for switch 1? Format your answer as letter,number or 'n/a'
3. At t=1, what is the source entry for switch 2? Format your answer as letter,number or 'n/a'
4. At t=1, what is the destination entry for switch 2? Format your answer as letter,number or 'n/a'
5. At t=2, what is the source entry for switch 1? Format your answer as letter,number or 'n/a'
6. At t=2, what is the destination entry for switch 1? Format your answer as letter,number or 'n/a'
7. At t=2, what is the source entry for switch 2? Format your answer as letter,number or 'n/a'
8. At t=2, what is the destination entry for switch 2? Format your answer as letter,number or 'n/a'
9. At t=3, what is the source entry for switch 1? Format your answer as letter,number or 'n/a'
10. At t=3, what is the destination entry for switch 1? Format your answer as letter,number or 'n/a'
11. At t=3, what is the source entry for switch 2? Format your answer as letter,number or 'n/a'
12. At t=3, what is the destination entry for switch 2? Format your answer as letter,number or 'n/a'
13. At t=4, what is the source entry for switch 1? Format your answer as letter,number or 'n/a'
14. At t=4, what is the destination entry for switch 1? Format your answer as letter,number or 'n/a'
15. At t=4, what is the source entry for switch 2? Format your answer as letter,number or 'n/a'
16. At t=4, what is the destination entry for switch 2? Format your answer as letter,number or 'n/a'

SOLUTION

1. At time t=1, (B,2) is added as an entry to switch table 1.
2. At time t=1, (A,1) is added as an entry to switch table 1.
3. At time t=1, (B,8) is added as an entry to switch table 2.
4. At time t=1, switch table 2 doesn't observe this frame
5. At time t=2, (K,7) is added as an entry to switch table 1.
6. At time t=2, since the entry for computer B in switch table 1 already exists, no new table entry is made
7. At time t=2, (K,14) is added as an entry to switch table 2.
8. At time t=2, since the entry for computer B in switch table 2 already exists, no new table entry is made
9. At time t=3, since the entry for computer K in switch table 1 already exists, no new table entry is made
10. At time t=3, switch table 1 doesn't observe this frame
11. At time t=3, since the entry for computer K in switch table 2 already exists, no new table entry is made
12. At time t=3, (H,11) is added as an entry to switch table 2.
13. At time t=4, (I,7) is added as an entry to switch table 1.
14. At time t=4, (F,6) is added as an entry to switch table 1.
15. At time t=4, (I,12) is added as an entry to switch table 2.
16. At time t=4, (F,8) is added as an entry to switch table 2.

LEARNING SWITCHES - ADVANCED

Consider the LAN below consisting of 10 computers connected by two self-learning Ethernet switches. (You may want to re-read section 6.4.3 in the text). At t=0 the switch table entries for both switches are empty. At t = 1, 2, 3, 4, 5, 6, 7, 8, and 9, a source sends to a destination as shown below, and the destination replies immediately (well before the next time step).



MAC addr	interface	TTL
C	8	9
F	8	5
J	13	7
D	8	4
K	14	6
H	11	9

MAC addr	interface	TTL
C	3	9
A	1	1
F	6	5
J	7	5
D	4	8
B	2	8
K	7	6
H	7	9

QUESTION LIST

1. At t=3, what two nodes communicated? Write your answer in alphabetical order as x,y (If there is only enough information for 1 node, write that, and if there's no information, write 'n/a')

2. At t=5, what two nodes communicated? Write your answer in alphabetical order as x,y (If there is only enough information for 1 node, write that, and if there's no information, write 'n/a')

3. At t=7, what two nodes communicated? Write your answer in alphabetical order as x,y (If there is only enough information for 1 node, write that, and if there's no information, write 'n/a')

4. At t=2, what two nodes communicated? Write your answer in alphabetical order as x,y (If there is only enough information for 1 node, write that, and if there's no information, write 'n/a')

SOLUTION

1. The transmission at t=3 was F-->J, so the answer is n/a.

2. The transmission at t=5 was F-->J, so the answer is F,J.

3. The transmission at t=7 was J-->H, so the answer is J.

4. The transmission at t=2 was C-->F, so the answer is n/a.

CDMA - ADVANCED

This time, assume there are 2 senders whom interfere with eachother and that the interfering transmitted bit signals are additive. The value received at a receiver, however, is now the sum of the transmitted bits from all senders.

Assume that sender 1 has a CDMA code of (-1, -1, 1, -1, -1, 1, 1, -1, -1) and sender 2 has a CDMA code of (-1, 1, -1, -1, -1, 1, -1, 1, -1) and their combined output is: (2, 0, 0, 2, 2, -2, 0, 0, 2)

QUESTION LIST

1. Assuming you are receiver 1, what is the decoded bit? If it can't be done, answer with 'n/a'

2. Assuming you are receiver 2, what is the decoded bit? If it can't be done, answer with 'n/a'

SOLUTION

1. Because the CDMA codes weren't well chosen, interference will render the receiver unable to decode the original bit

2. Because the CDMA codes weren't well chosen, interference will render the receiver unable to decode the original bit

CDMA - BASIC

Imagine a scenario where a sender and receiver using CDMA (Section 7.2.1 from the text) are exchanging data. Assume both the sender and receiver use a CDMA code of size 9, where M = -1, -1, 1, 1, 1, -1, -1, -1, 1.

QUESTION LIST

1. Given the CDMA code above and the bit d=1, what is the encoded output? Separate each value with a comma and no spaces

2. Given the CDMA code above and the output string: -1,-1,1,1,1,-1,-1,-1,1, what is the decoded bit value?

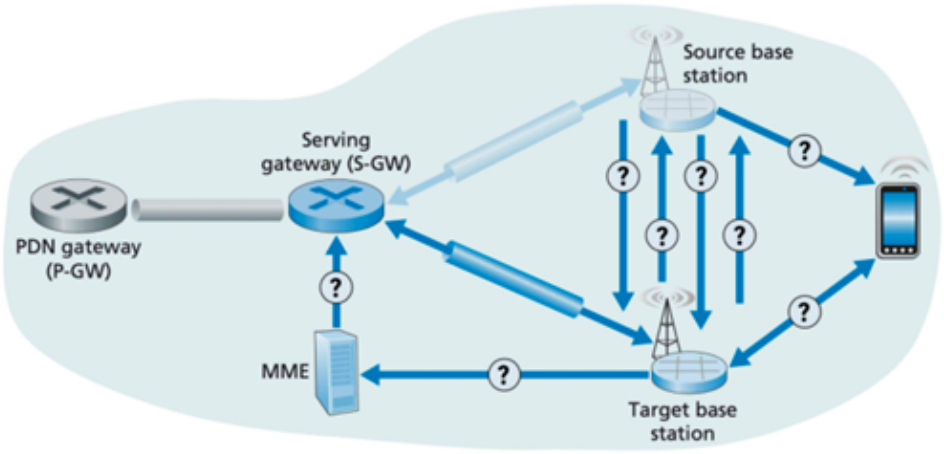
SOLUTION

1. The encoded output is: -1,-1,1,1,1,-1,-1,-1,1

2. The original bit value is: 1

4G WIRELESS HANDOVER

Consider the scenario, where a mobile phone connected to a base station (source) is streaming a video from the internet, and the mobile phone is going to switch to a closer base station (target) without interrupting their video stream. This process (called a handover) is shown in the diagram below.



Take a look at the statements below and put them in the correct order.

A. The source base station receives the Handover Request Acknowledgement message and informs the mobile device of the target base station's identity and channel access information. At this point, the mobile device can begin send- ing/receiving datagrams to/from the new target base station. From the mobile device's point of view, handover is now complete! However, there is still a bit of work to be done within the network.

B. The current (source) base station selects the target base station, and sends a Handover Request message to the target base station.

C. The target base station informs the MME that it (the target base station) will be the new base station servicing the mobile device. The MME, in turn, signals to the Serving Gateway and the target base station to reconfigure the Serving- Gateway-to-base-station tunnel to terminate at the target base station, rather than at the source base station.

D. At this point, the target base station can also begin delivering datagrams to the mobile device, including datagrams forwarded to the target base station by the source base station during handover, as well as datagrams newly arriving on the reconfigured tunnel from the Serving Gateway. It can also forward outgo- ing datagrams received from the mobile device into the tunnel to the Serving Gateway.

E. The target base station confirms back to the source base station that the tunnel has been reconfigured, allowing the source base station to release resources associated with that mobile device.

F. The target base station checks whether it has the resources to support the mobile device and its quality of service requirements. If so, it pre-allocates channel resources (e.g., time slots) on its radio access network and other resources for that device. This pre-allocation of resources frees the mobile device from having to go through the time-consuming base-station association protocol discussed earlier, allowing handover to be executed as fast as possible. The target base station replies to the source base station with a Handover Request Acknowledge message, containing all the information at the target base station that the mobile device will need to associate with the new base station.

G. The source base station will also stop forwarding datagrams to the mobile device and instead forward any tunneled datagrams it receives to the target base station, which will later forward these datagrams to the mobile device.

QUESTION LIST

1. Which lettered step (A-G) above corresponds to the first step in the handover process?

2. Which lettered step (A-G) above corresponds to the second step in the handover process?

3. Which lettered step (A-G) above corresponds to the third step in the handover process?

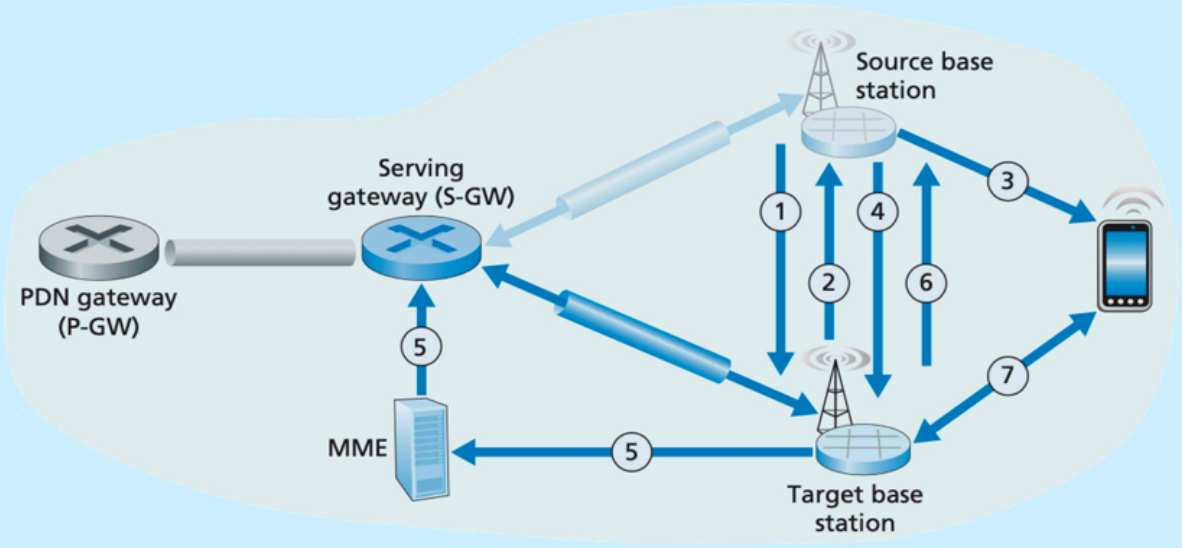
4. Which lettered step (A-G) above corresponds to the fourth step in the handover process?

5. Which lettered step (A-G) above corresponds to the fifth step in the handover process?

6. Which lettered step (A-G) above corresponds to the sixth step in the handover process?

7. Which lettered step (A-G) above corresponds to the seventh and final step in the handover process?

SOLUTION



1. The first step corresponds to B

2. The second step corresponds to F

3. The third step corresponds to A

4. The fourth step corresponds to G

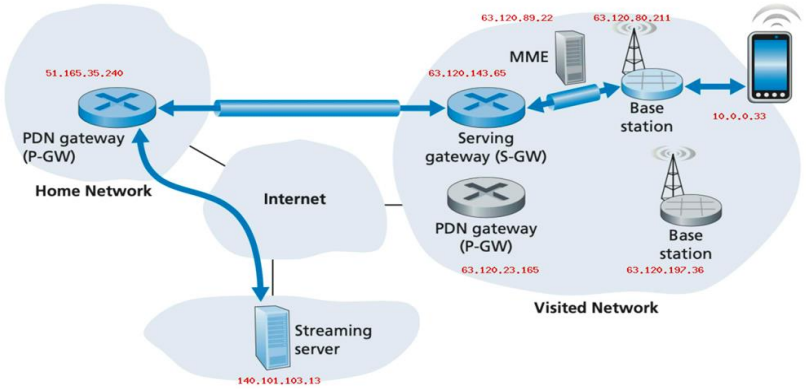
5. The fifth step corresponds to C

6. The sixth step corresponds to E

7. The seventh and final step corresponds to D

4G WIRELESS TUNNELING

In the graphic below, a mobile phone has a TCP connection with the streaming server using wireless 4G. Assume that both the mobile phone and the server use TCP port 11556 and any intermediary nodes that tunnel the datagrams use port 5678.



QUESTION LIST

1. What is the source IP address of the datagram sent from the mobile phone to the base station?
2. What is the destination IP address of the datagram sent from the mobile phone to the base station?
3. What is the transport-layer port number in the datagram sent from the mobile phone to the base station?
4. What transport-layer protocol is indicated in the datagram sent from the mobile phone to the base station?
5. Is there an encapsulated datagram within the datagram sent from the mobile phone to the base station? Answer Yes or No
6. What is the source IP address of the datagram sent from the base station to the serving gateway?
7. What is the destination IP address of the datagram sent from the base station to the serving gateway?
8. What is the transport-layer port number in the datagram sent from the base station to the serving gateway?
9. What transport-layer protocol is indicated in the datagram sent from the base station to the serving gateway?
10. Is there an encapsulated datagram within the datagram sent from the base station to the serving gateway? Answer Yes or No
11. What is the source IP address of the encapsulated datagram?
12. What is the destination IP address of the encapsulated datagram?
13. What is the transport-layer port number in the encapsulated datagram?
14. What transport-layer protocol is indicated in the encapsulated datagram?
15. What is the source IP address of the datagram sent from the serving gateway to the PDN gateway?
16. What is the destination IP address of the datagram sent from the serving gateway to the PDN gateway?
17. What is the transport-layer port number in the datagram sent from the serving gateway to the PDN gateway?
18. What transport-layer protocol is indicated in the datagram sent from the serving gateway to the PDN gateway?
19. Is there an encapsulated datagram within the datagram sent from the serving gateway to the PDN gateway? Answer Yes or No
20. What is the source IP address of the encapsulated datagram?
21. What is the destination IP address of the encapsulated datagram?
22. What is the transport-layer port number in the encapsulated datagram?
23. What transport-layer protocol is indicated in the encapsulated datagram?
24. What is the source IP address of the datagram sent from the PDN gateway to the server?
25. What is the destination IP address of the datagram sent from the PDN gateway to the server?
26. What is the transport-layer port number in the datagram sent from the PDN gateway to the server?
27. What transport-layer protocol is indicated in the datagram sent from the PDN gateway to the server?
28. Is there an encapsulated datagram within the datagram sent from the PDN gateway to the server? Answer Yes or No
29. Is data encrypted between the mobile phone and the base station? Answer Yes or No
30. Does the server know about any of the wireless tunneling? Answer Yes or No

SOLUTION

1. From the mobile phone to the base station, the source address is 10.0.0.33

2. From the mobile phone to the base station, the destination address is 140.101.103.13

3. From the mobile phone to the base station, the port used is 11556

4. From the mobile phone to the base station, the protocol used is TCP

5. Is there an encapsulated datagram? No

6. From the base station to the serving gateway, the source address is 63.120.80.211

7. From the base station to the serving gateway, the destination address is 63.120.143.65

8. From the base station to the serving gateway, the port used is 5678

9. From the base station to the serving gateway, the protocol used is UDP

10. Is there an encapsulated datagram? Yes

11. The source address of the encapsulated datagram is 10.0.0.33

12. The destination address of the encapsulated datagram is 140.101.103.13

13. The port used in the encapsulated datagram is 11556

14. The protocol used in the encapsulated datagram is TCP

15. From the serving gateway to the PDN gateway, the source address is 63.120.143.65

16. From the serving gateway to the PDN gateway, the destination address is 51.165.35.240

17. From the serving gateway to the PDN gateway, the port used is 5678

18. From the serving gateway to the PDN gateway, the protocol used is UDP

19. Is there an encapsulated datagram? Yes

20. The source address of the encapsulated datagram is 10.0.0.33

21. The destination address of the encapsulated datagram is 140.101.103.13

22. The port used by the encapsulated datagram is 11556

23. The protocol used by the encapsulated datagram is TCP

24. From the PDN gateway to the server, the source address is 51.165.35.240

25. From the PDN gateway to the server, the destination address is 140.101.103.13

26. From the PDN gateway to the server, the port used is 11556

27. From the PDN gateway to the server, the protocol used is TCP

28. Is there an encapsulated datagram? No

29. Yes, the base station and mobile phone encrypt data between them

30. No, the server is blissfully unaware of the wireless infrastructure or tunneling